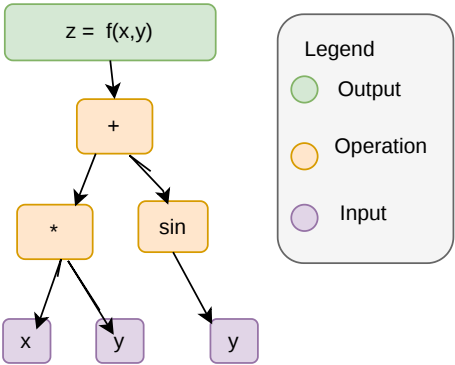


Computation Graph vs Instruction Tape (Reverse-Mode AD)

Computation Graph



Pros

- Intuitive
- Easy to Visualize
- Good for Forward Mode

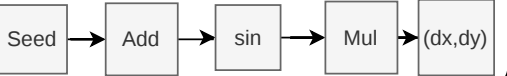
Cons

- Higher memory Overhead
- Pointer-heavy
- Expensive to traverse

Instruction Tape (Topological Order)

Step	Ops	Inputs	Outputs	Stored?
1	Load	x	v1	Yes
2	Load	y	v2	Yes
3	Mul	v1,v2	v3	Yes
4	Sin	v2	v4	Yes
5	Add	v3,v4	v5 (z)	Yes

Reverse Sweep (BP)



Pros

- Cache-friendly
- Reusable across rows
- Linear traversal, no pointer chasing
- Compiled once at bind time

Cons

- Harder to Visualize
- Requires topological order
- Build cost (once)